

URMI, Russian Federation

WMO: 316240

Lat: 49.400N

Lon: 133.233E

Elev: 201

StdP: 98.93

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.7	(c) -34.5	(d) -39.1	(e) 0.1	(f) -36.4	(g) -37.0	(h) 0.1	(i) -34.1	(j) 5.4	(k) -19.5	(l) 5.1	(m) -18.5	(n) 0.0	(o) 340	(p) 0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.5	(c) 29.7	(d) 21.5	(e) 27.7	(f) 20.5	(g) 26.1	(h) 19.5	(i) 23.1	(j) 27.6	(k) 22.0	(l) 25.8	(m) 21.0	(n) 24.7	(o) 1.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	16.8	25.4	20.7	15.7	24.2	19.7	14.8	23.1	70.1	27.7	65.8	25.9	62.1	24.7	33.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.4	(b) 4.8	(c) 3.4	(e) -38.6	(f) 33.8	(g) 2.0	(h) 2.6	(i) -40.1	(j) 35.7	(k) -41.3	(l) 37.2	(m) -42.4	(n) 38.7	(o) -43.9	(p) 40.6		
			DB	WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-22.5	-18.2	-8.1	2.2	10.3	16.3	19.8	18.2	11.3	1.7	-11.5	-21.8		
	DBStd	15.76	4.82	4.62	5.81	4.04	3.75	3.26	2.50	2.92	3.86	4.26	6.33	5.04		
	HDD10.0	4557	1007	789	563	236	43	1	0	1	29	258	646	987		
	HDD18.3	6838	1266	1022	821	483	250	78	14	37	211	516	896	1245		
	CDD10.0	875	0	0	0	2	52	189	306	256	69	1	0	0		
	CDD18.3	113	0	0	0	0	1	16	61	34	2	0	0	0		
	CDH23.3	1191	0	0	0	4	78	297	517	272	24	0	0	0		
	CDH26.7	298	0	0	0	0	14	87	149	45	3	0	0	0		
	WSAvg	0.9	0.7	0.9	1.2	1.1	1.0	0.7	0.6	0.6	0.8	1.1	0.9	0.8		
	PrecAvg	818	7	8	21	41	86	111	170	168	105	46	20	13		
(15) Precipitation	PrecMax	1063	32	36	62	121	193	230	314	251	247	116	74	37		
	PrecMin	585	0	0	2	4	13	26	27	38	19	5	1	1		
	PrecStd	140	8	8	16	31	46	59	68	56	49	30	17	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.9	-1.4	10.7	23.2	28.5	32.5	33.1	30.0	26.0	18.8	5.1	-4.8		
		MCWB	-7.4	-4.5	3.7	12.0	16.0	22.1	22.8	23.0	19.5	11.9	1.8	-5.6		
	2%	DB	-9.3	-4.1	6.9	17.4	25.7	29.0	30.2	27.9	23.2	14.9	2.2	-8.3		
		MCWB	-10.5	-6.5	1.5	8.7	14.3	20.3	22.2	21.8	17.1	8.5	0.0	-9.3		
	5%	DB	-11.8	-6.6	4.6	14.1	22.8	26.7	28.1	26.2	21.1	11.9	0.3	-10.8		
		MCWB	-12.8	-8.4	0.0	6.9	12.8	18.8	21.5	20.7	15.6	6.7	-1.0	-11.7		
	10%	DB	-14.1	-8.8	2.1	11.2	20.1	24.5	26.2	24.6	19.0	9.4	-1.5	-13.3		
		MCWB	-14.9	-10.3	-1.3	4.9	11.9	18.0	20.7	19.9	14.6	5.5	-3.2	-14.1		
	0.4%	WB	-7.3	-4.1	3.9	13.1	17.3	24.2	24.7	24.4	20.7	12.4	2.0	-5.5		
		MCDB	-6.1	-1.5	10.2	22.0	26.1	31.0	29.5	28.2	25.1	17.5	3.8	-4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-10.4	-6.4	1.8	9.3	15.5	21.5	23.2	23.0	18.6	9.7	0.5	-9.1		
		MCDB	-9.3	-4.3	6.1	16.0	22.7	26.5	27.9	26.7	21.8	13.1	1.9	-8.3		
	5%	WB	-12.8	-8.3	0.4	7.3	14.1	19.9	22.2	21.7	16.8	7.6	-1.0	-11.7		
		MCDB	-11.8	-6.5	3.9	13.5	20.6	24.9	26.3	24.8	19.3	10.8	0.4	-10.8		
	10%	WB	-14.9	-10.3	-1.0	5.4	12.7	18.7	21.4	20.6	15.2	6.0	-3.2	-14.1		
		MCDB	-14.2	-8.8	2.2	10.2	18.4	23.5	25.3	23.5	18.0	9.0	-1.6	-13.3		
	5% DB	MDBR	15.2	17.6	16.7	14.3	15.1	13.8	11.5	11.7	14.3	13.2	12.3	13.4		
		MCDBR	16.1	18.6	17.3	20.7	22.3	19.2	16.2	15.0	18.2	18.4	12.8	13.6		
		MCWBR	14.9	16.3	12.2	12.8	12.0	10.9	9.2	9.3	11.7	12.1	10.4	12.5		
		MCWBR	16.0	18.3	16.2	19.0	18.7	15.7	13.5	12.0	14.4	16.3	12.2	13.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	14.8	16.0	11.6	12.2	10.9	9.7	8.4	8.1	10.1	11.4	10.2	12.3		
	taub		0.275	0.312	0.384	0.519	0.559	0.509	0.495	0.433	0.397	0.407	0.339	0.282		
		taud	2.186	2.065	1.924	1.754	1.774	2.001	2.094	2.272	2.287	2.070	2.111	2.152		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		763	809	800	725	718	763	768	803	794	684	653	686		
		Edh at Noon	77	112	155	206	211	169	152	122	108	112	84	69		
	RadAvg		1.78	2.99	4.40	4.65	4.96	5.35	4.91	4.51	3.68	2.48	1.66	1.38		
(44) All-Sky Solar Radiation	RadStd		0.04	0.07	0.32	0.49	0.49	0.77	0.47	0.40	0.38	0.21	0.14	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page